Project Design Phase-I
Solution Architecture

Date	8 March 2026
Team ID	699d5fa0c40b5ad0e9ee5a7d
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Solution Architecture:

The proposed system is designed to predict hypertension using a machine learning-based web application. The architecture consists of multiple components that work together to collect input data, process it, and generate predictions.

First, the user (doctor or patient) enters the required health parameters through a web interface developed using HTML and CSS. These inputs are sent to the backend server implemented using the Flask framework.

The Flask backend processes the input data and performs data preprocessing such as scaling and formatting to ensure the data is suitable for the machine learning model. The processed data is then passed to the trained machine learning model built using Scikit-learn.

The machine learning model analyzes the patient's health parameters and predicts the blood pressure category or hypertension stage. Finally, the prediction result is returned to the Flask server and displayed on the web interface for the user.

This architecture enables efficient data processing, accurate prediction, and easy interaction through a simple web-based interface, making the system useful for healthcare monitoring and early detection of hypertension.

Example - Solution Architecture Diagram:

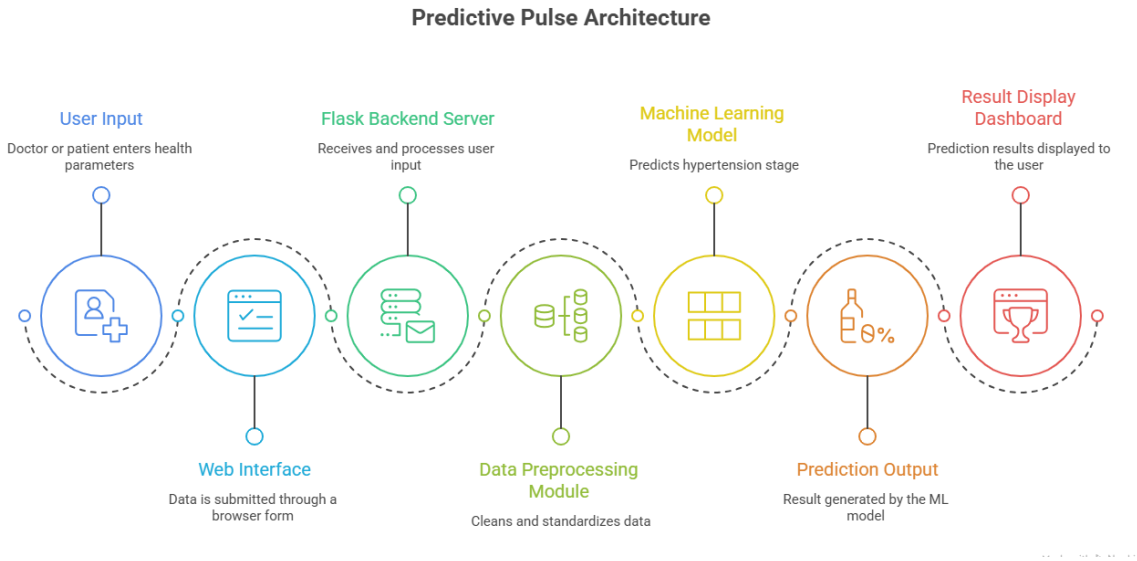

Figure 1: Architecture and data flow of the voice patient diary sample application